

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the integration of different avionic system.
- Q.22 Describe the process of identification of electrical diagram.
- Q.23 What is various common mode of failures in Flight Control system.
- Q.24 Explain the structure of stable inertial platform.
- Q.25 Differentiate between volatile and non-volatile members?
- Q.26 What are the salient features of FBW system?
- Q.27 Demonstrate the working of instrument landing system.
- Q.28 What are different control laws?
- Q.29 How does Fire Control System Work?
- Q.30 Describe failure survival system.
- Q.31 Explain a flight control system
- Q.32 What is dead reckoning navigation system and explain any one in detail.
- Q.33 What is the need of avionic systems?
- Q.34 Describe integration process of aircraft systems.
- Q.35 What is the application of Power utilization to avionic?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain the common mode of failures in Flight Control system. Explain inertial alignment as well.
- Q.37 Draw and explain the working of component of avionic subsystem.
- Q.38 Explain the functioning of
 - a) Cathode Ray tube b) Light Emitting Diode
 - c) Radar

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Roll No.

6th Sem / AME Sub : Aircraft Avionics

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which avionics subsystem is responsible for controlling the movement of aircraft surfaces like ailerons, elevators and rudders?
 - a) Communication system
 - b) Fight Management System (FMS)
 - c) Flight Control System
 - d) Navigation System
- Q.2 In aircraft electronic subsystems, what does the term "EFIS" stand for?
 - a) Engine Fuel Injection System
 - b) Electrical Flight Information System
 - c) Enhanced Flight Instrumentation System
 - d) Emergency Fire Ignition System
- Q.3 What is the purpose of a digital data bus in a computer system?
 - a) To provide power to the CPU
 - b) To transmit data between different components of the computer system
 - c) To cool down the system components
 - d) To display graphics on the monitor

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- Q.4 Which type of memory device retains its data even when the power is turned off and is commonly used to store firmware and system settings?
- SRAM
 - DRAM
 - EEPROM
 - ROM
- Q.5 What is the primary function of an Airborne Collision Avoidance System (ACAS) in radar technology?
- To provide real-time traffic updates to air traffic controllers
 - To detect and track enemy aircraft
 - To monitor the aircraft's engine performance
 - To prevent mid-air collisions by providing advisories to pilots
- Q.6 What is the primary function of an Electronic Flight Control System (EFCS) in modern aircraft?
- Monitoring passenger entertainment systems
 - Providing real-time weather updates to the crew
 - Automatically adjusting flight control surface to maintain stable flight
 - Controlling cabin lighting and temperature
- Q.7 What is the primary purpose of the Primary Flight Display (PFD) in an aircraft cockpit?
- To display weather radar information
 - To provide information about engine performance
 - To show essential flight parameters such as altitude, airspeed and attitude
 - To control the aircraft's autopilot system
- Q.8 Which of the following cockpit instruments provides information about the aircraft's heading relative to magnetic north?
- Vertical Speed Indicator (VSI)
 - Attitude Indicator (AI)

- Magnetic Compass
 - Altimeter
- Q.9 What is the role of avionics software in the integration process?
- To manufacture physical components for avionics systems
 - To provide real time monitoring and control of aircraft systems
 - To design aircraft structures and aerodynamics
 - To conduct safety inspections of avionics systems
- Q.10 Which phase of aircraft development typically involves extensive testing of integrated avionics systems to ensure functionality and reliability?
- Design phase
 - Manufacturing phase
 - Testing & certification phase
 - Operational phase

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What is the function of oscillator?
- Q.12 What is the importance of compass swing?
- Q.13 Name the important memory devices.
- Q.14 What is fly by wire?
- Q.15 What is the role of Fiber Optics?
- Q.16 What is redundancy?
- Q.17 Where is avionics used in military aircraft?
- Q.18 What is the use of DVI?
- Q.19 What is the use of Copper Harper scale?
- Q.20 Name the different flight controls?